THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapters 8 & 9: **Introduction to Trigonometry, Some Applications of Trigonometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{2}$	$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}$
2	**(a)** 45°	$\cot 45^\circ = 1$
3	**(c)** $\frac{1}{2}$	$\frac{3}{4} - \frac{1}{4}$
4	**(c)** Incorrect — $\tan 90^\circ$ is not defined	$\cos 90^\circ = 0$
5	**(a)** Student A is correct	$A = 30^\circ$
6	**(d)** 6 m	$\frac{3}{\sin 30^\circ}$
7	**(d)** 45°	$\tan\theta = 1$
8	**(c)** $100\sqrt{3}$ m	$\frac{100}{\tan 30^\circ}$
9	**(c)** Incorrect — it lies between 0° and 90°	Measured from the horizontal
10	**(b)** Student A is correct	$\frac{50}{\sin 30^\circ}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\cos\theta = \frac{15}{17}$, $\tan\theta = \frac{8}{15}$	1+1
12	$4 - 4 + \frac{3}{4} + 0 = \frac{3}{4}$	2
13	$\frac{50\sqrt{3}}{\sqrt{3}} = 50$ m	1+1
14	$10\tan 60^\circ = 10\sqrt{3} \approx 17.32$ m	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$= \frac{(\sin A+\cos A-1)(\sin A+\cos A+1)}{\sin A\cos A} = \frac{(\sin A+\cos A)^2-1}{\sin A\cos A} = \frac{2\sin A\cos A}{\sin A\cos A} = 2$	1+1+1
16	Building 20 m; total $20\sqrt{3}$ ⇒ flagpole $= 20(\sqrt{3}-1) \approx 14.64$ m	1+1+1
17	$\sin A = 1 - \sin^2 A = \cos^2 A \Rightarrow \cos^2 A + \cos^4 A = \sin A + \sin^2 A = 1$	1+1+1
	OR: $2(\sin^2\theta + \cos^2\theta) = 2$	1+1+1
18	$30\sqrt{3} - \frac{30}{\sqrt{3}} = 20\sqrt{3} \approx 34.64$ m	1+1+1
	OR: $\frac{75}{\sin 60^\circ} = \frac{150}{\sqrt{3}} = 50\sqrt{3} \approx 86.6$ m	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$= \frac{\sin A(1-2\sin^2 A)}{\cos A(2\cos^2 A-1)}$; $1 - 2\sin^2 A = 2\cos^2 A - 1 \Rightarrow \tan A$ (3). $A = 30^\circ$: $\frac{\frac{1}{2}-\frac{1}{4}}{\frac{3\sqrt{3}}{4}-\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$ ✓ (2)	3+2
	OR: Write $1 = \sec^2 A - \tan^2 A$ in the numerator: $\frac{(\sec A+\tan A)(1-\sec A+\tan A)}{\tan A-\sec A+1} = \sec A + \tan A = \frac{1+\sin A}{\cos A}$	2+2+1
20	(ii) 15 m (45°) (iii) $15 + 15\tan 30^\circ = 15 + 5\sqrt{3} \approx 23.66$ m	1+2+2
	OR: $\frac{H-60}{d} = \frac{1}{\sqrt{3}}$, $\frac{H+60}{d} = \sqrt{3} \Rightarrow H + 60 = 3(H - 60) \Rightarrow H = 120$ m	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 8 & 9: **Introduction to Trigonometry, Some Applications of Trigonometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1	$\frac{1}{\sqrt{3}} \cdot \sqrt{3}$
2	**(b)** $\frac{5}{3}$	Reciprocal
3	**(d)** 1	$\sin^2 A + \cos^2 A$
4	**(c)** Incorrect — $\frac{\sqrt{3}}{2} \neq 1$	$\sin 2A \neq 2 \sin A$
5	**(a)** Student A is correct	$\frac{1}{\sin 30^\circ}$
6	**(d)** 60°	$\tan\theta = \sqrt{3}$
7	**(c)** 7.5 m	Angle with ground 30°: $15 \sin 30^\circ$
8	**(a)** 50 m	$\tan 45^\circ = 1$
9	**(b)** Incorrect — $\tan\theta$ halves, not θ	$\tan\theta = \frac{h}{s}$
10	**(b)** Student A is correct	$\tan 60^\circ = \frac{h}{d}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Divide by $\cos\theta$: $\frac{5\tan\theta - 3}{5\tan\theta + 2} = \frac{4-3}{4+2} = \frac{1}{6}$	1+1
12	$\theta = 30^\circ \Rightarrow \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$	1+1
13	$30\sqrt{3} \approx 51.96$ m	2
14	$20\sqrt{3} \approx 34.64$ m	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$= \frac{\cos^2 A}{\cos A - \sin A} - \frac{\sin^2 A}{\cos A - \sin A} = \frac{(\cos A - \sin A)(\cos A + \sin A)}{\cos A - \sin A}$	1+1+1
16	$h\sqrt{3} - \frac{h}{\sqrt{3}} = 30 \Rightarrow h = 15\sqrt{3} \approx 25.98$ m	1+1+1
17	$\tan^2\theta = \sec^2\theta - 1 = \left(x - \frac{1}{4x}\right)^2 \Rightarrow \tan\theta = \pm\left(x - \frac{1}{4x}\right) \Rightarrow$ sum $2x$ or $\frac{1}{2x}$	1+1+1
	OR: $\sin^2\theta\left(\frac{1}{\cos^2\theta} - 1\right) = \sin^2\theta\tan^2\theta$	1+1+1
18	Girl: 75 m; boy: $50\sqrt{2} \approx 70.71$ m ⇒ girl's, by about 4.29 m	1+1+1
	OR: $60\sqrt{3} + 60 = 60(\sqrt{3} + 1) \approx 163.92$ m	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Divide numerator and denominator by $\cos A$: $\frac{\frac{1}{\sin A} - 1}{\frac{1}{\sin A} + 1}$	2+2+1
	OR: LHS $= \csc^2 A \sec^2 A = \frac{1}{\sin^2 A \cos^2 A} = \frac{1}{\sin^2 A(1 - \sin^2 A)}$	2+2+1
20	(i) $\frac{200}{\sqrt{3}} \approx 115.47$ m (ii) $200\sqrt{3} \approx 346.41$ m (iii) ≈ 230.94 m (iv) ≈ 115.47 m/min	1+1+2+1
	OR: $100\sqrt{3} - 100 = 100(\sqrt{3} - 1) \approx 73.2$ m; speed ≈ 7.32 m/s	3+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 8 & 9: **Introduction to Trigonometry, Some Applications of Trigonometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1	$0 + 1$
2	**(d)** 60°	$\tan 60^\circ$
3	**(a)** $\frac{\sqrt{3}}{2}$	$2 \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2}$
4	**(c)** Incorrect — base < hypotenuse	$0 < \cos A < 1$
5	**(a)** Student A is correct	Identity
6	**(c)** $\frac{20}{\sqrt{3}}$ m	$\frac{20}{\tan 60^\circ}$
7	**(a)** 30°	$\tan\theta = \frac{1}{\sqrt{3}}$
8	**(a)** 50 m	$\tan 45^\circ = 1$
9	**(a)** Incorrect — shadows become shorter	Larger θ ⇒ smaller $\frac{h}{\tan\theta}$
10	**(b)** Student A is correct	$\frac{3000}{\sqrt{3}}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$BC = 12$; $\sin A = \frac{12}{13}$, $\tan C = \frac{5}{12}$	1+1
12	$\frac{1}{3}$	2
13	$\frac{4}{\cos 45^\circ} = 4\sqrt{2} \approx 5.66$ m	2
14	$30 \tan 30^\circ = 10\sqrt{3} \approx 17.32$ m	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sec^2\theta + \csc^2\theta = \frac{1}{\sin^2\theta\cos^2\theta}$; $\tan\theta + \cot\theta = \frac{1}{\sin\theta\cos\theta}$ ✓	1+1+1
16	$60\sqrt{3} + 20\sqrt{3} = 80\sqrt{3} \approx 138.56$ m	1+1+1
17	$\sin\theta = (\sqrt{2}-1)\cos\theta \Rightarrow \cos\theta = (\sqrt{2}+1)\sin\theta \Rightarrow \cos\theta - \sin\theta = \sqrt{2}\sin\theta$	1+1+1
	OR: $\frac{\sin A}{\tan A} = \frac{m}{n} \cdot \frac{\sin B}{\tan B} \Rightarrow \cos B = \frac{n}{m}\cos A$; with $\sin B = \frac{\sin A}{m}$: $\frac{\sin^2 A}{m^2} + \frac{n^2\cos^2 A}{m^2} = 1 \Rightarrow$ result	1+1+1
18	$100\sqrt{3} - \frac{100}{\sqrt{3}} = \frac{200}{\sqrt{3}} \approx 115.47$ m	1+1+1
	OR: $h = x\sqrt{3} = \frac{x+30}{\sqrt{3}} \Rightarrow x = 15$ m; $h = 15\sqrt{3} \approx 25.98$ m	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$(\sin A + \cos A) \cdot \frac{1}{\sin A\cos A} = \frac{1}{\cos A} + \frac{1}{\sin A}$ (3). $A = 45^\circ$: $\sqrt{2} \times 2 = 2\sqrt{2} = \sqrt{2} + \sqrt{2}$ ✓ (2)	3+2
	OR: $x^2 - y^2 = a^2(\sec^2\theta - \tan^2\theta) + b^2(\tan^2\theta - \sec^2\theta) = a^2 - b^2$ (cross terms cancel)	2+2+1
20	(i) $\frac{48}{\sqrt{3}} = 16\sqrt{3} \approx 27.71$ m (ii) $16\sqrt{3} \times \frac{1}{\sqrt{3}} = 16$ m (iii) 64 m	2+2+1
	OR: $h = x$, $\sqrt{3}h = x + 1000 \Rightarrow h = \frac{1000}{\sqrt{3}-1} = 500(\sqrt{3}+1) \approx 1366$ m	2+2+1

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ANSWER KEY & MARKING SCHEME — SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	$\frac{1}{\cos 60^\circ}$
2	**(a)** $\frac{4}{5}$	Sides 3, 4, 5
3	**(b)** 0	Same terms
4	**(a)** Incorrect — it means $(\sin\theta)^2$	Notation
5	**(b)** Student A is correct	$2-1$
6	**(d)** 10 m	$20\cos 60^\circ$
7	**(d)** $5\sqrt{3}$ m	$\frac{5}{\tan 30^\circ}$
8	**(d)** 40 m	$80\sin 30^\circ$
9	**(c)** Correct	Definition
10	**(a)** Student A is correct	$\frac{30}{\sqrt{3}}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$4\sin^2\theta+3=4 \Rightarrow \sin\theta=\frac{1}{2} \Rightarrow \theta=30^\circ$, $\tan\theta=\frac{1}{\sqrt{3}}$	1+1
12	$\frac{1/\sqrt{2}}{\frac{2}{\sqrt{3}}+2}=\frac{3\sqrt{2}-\sqrt{6}}{8}$	1+1
13	$\frac{6}{\sqrt{3}}=2\sqrt{3}\approx 3.46$ m	2
14	$1000\sqrt{3}\approx 1732$ m	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$(\sin^2 A-\cos^2 A)(\sin^2 A+\cos^2 A)=1-\cos^2 A-\cos^2 A$	1+1+1
16	Width 12 m; height above window $12\sqrt{3} \Rightarrow 12(1+\sqrt{3})\approx 32.78$ m	1+1+1
17	Multiply by $\frac{1-\sin A}{1-\sin A}$: $\frac{\cos A(1-\sin A)}{\cos^2 A}=\frac{1-\sin A}{\cos A}$	1+1+1
	OR: $l=\frac{\cos^2\theta}{\sin\theta}$, $m=\frac{\sin^2\theta}{\cos\theta} \Rightarrow lm=\sin\theta\cos\theta$, $l^2+m^2+3=\frac{1}{\sin^2\theta\cos^2\theta} \Rightarrow$ product 1	1+1+1
18	$75\sqrt{3}-25\sqrt{3}=50\sqrt{3}\approx 86.6$ m	1+1+1
	OR: $\tan\theta=\frac{28.5}{28.5}=1 \Rightarrow 45^\circ$; for 60°: $\frac{28.5}{\sqrt{3}}\approx 16.45$ m	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\frac{1}{\sec A-\tan A}=\sec A+\tan A$ and $\frac{1}{\sec A+\tan A}=\sec A-\tan A \Rightarrow$ both sides equal $\tan A$	2+2+1
	OR: $2+2\tan^2 A=2\sec^2 A$ (3); $2\times 4=8$ (2)	3+2
20	(i) $2\sin 30^\circ=1$ km (ii) $2\cos 30^\circ=\sqrt{3}\approx 1.73$ km (iii) $\frac{2}{4}$ h $=30$ min	2+2+1
	OR: Distance flown $=20\sqrt{3}-20=20(\sqrt{3}-1)\approx 14.64$ m $\Rightarrow$ speed ≈ 7.32 m/s	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.